

Civ Eng 2200 Statics
Spring 2022 Exam 3 (for practice)

Section 1 – Internal Forces

1. (5 pt) What approach is typically applied when determining the internal loads within a rigid body?

- ✓ A. method of sections
- B. method of joints
- C. integration
- D. two-force member analysis
- E. particle equilibrium analysis

2. (5 pt) The force that acts perpendicular to the “sectioned” surface is called the ____.

- ✓ A. normal force
- B. bending force
- C. universal force of attraction
- D. shear force
- E. gravitational force
- F. centrifugal force

3. (5 pt) Determine the magnitude of the shear force at section $b-b$ that lies 0.1 m from A , located 0.1 m along crank arm OA , if $m = 5$ kg.

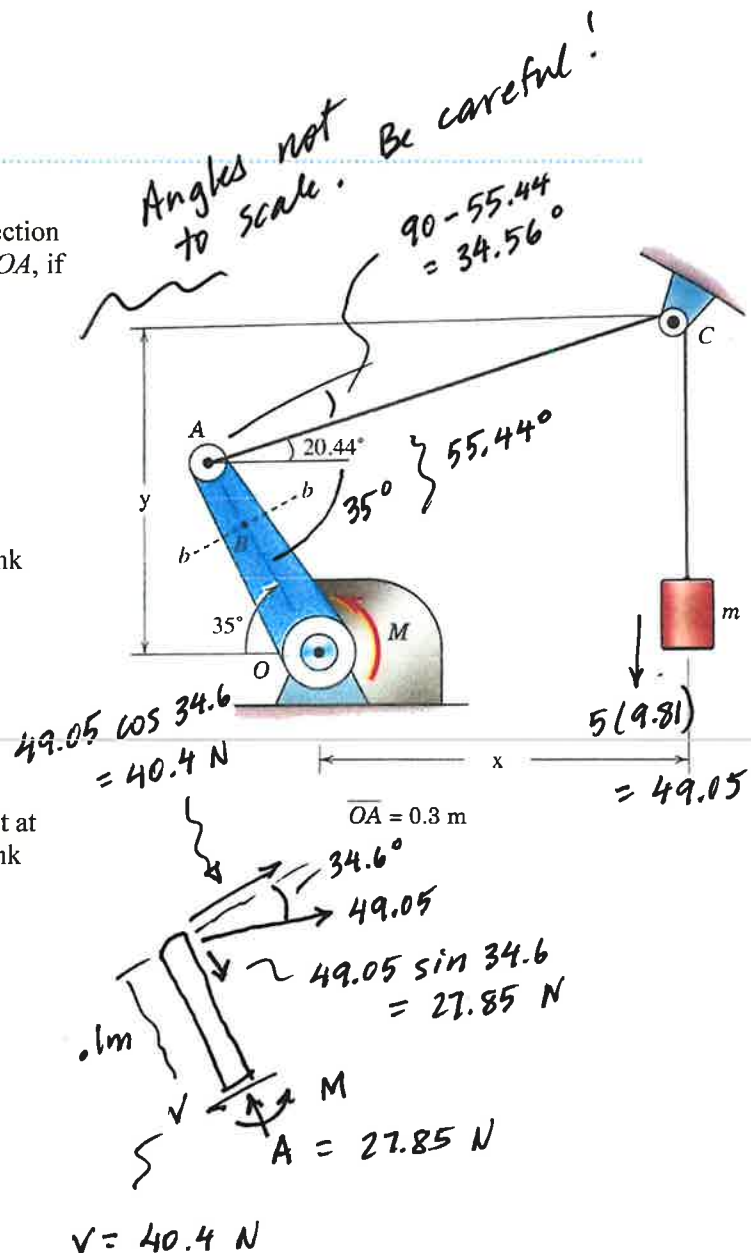
- ✓ A. 40.4 N
- B. 23.5 N
- C. 34.1 N
- D. 68.2 N
- E. 52.7 N

4. (5 pt) Determine the magnitude of the normal force at section $b-b$ that lies 0.1 m from A , located 0.1 m along crank arm OA , if $m = 5$ kg.

- ✓ A. 27.8 N
- B. 13.3 N
- C. 32.0 N
- D. 21.4 N
- E. 19.5 N

5. (5 pt) Determine the magnitude of the bending moment at section $b-b$ that lies 0.1 m from A , located 0.1 m along crank arm OA , if $m = 5$ kg.

- ✓ A. 4.04 N-m
- B. 4.79 N-m
- C. 3.26 N-m
- D. 3.01 N-m
- E. 2.38 N-m



$$\begin{aligned} \sum M_{cut} &= 0; \\ M - 40.4(0.1) &= 0 \\ M &= 4.04 \text{ N}\cdot\text{m} \end{aligned}$$

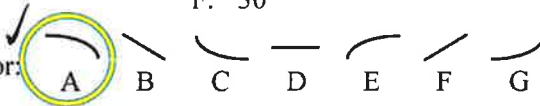
Section 2 – Shear and Moment Diagrams

Provide information for constructing the **shear diagram**. The x origin is on the left and the support reactions are $A_y = 61$ kN acting up, $B_y = 29$ kN acting up. A is a pin and B is a roller.

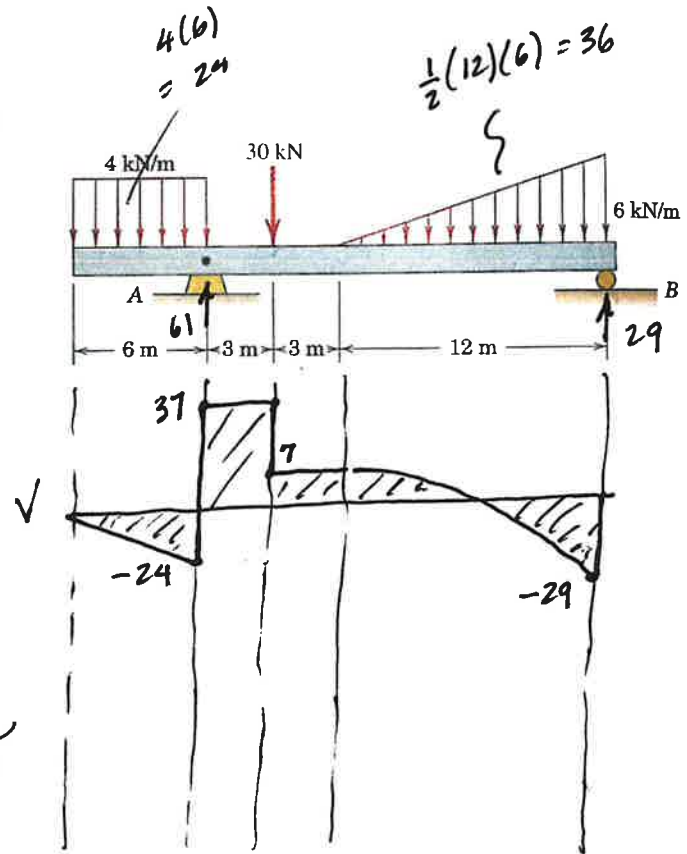
6. (3 pt) What happens to the shear diagram at $x = 6$ m?
- ✓ A. Increase 61 kN
 - B. Decrease 24 kN
 - C. Increase 24 kN
 - D. Decrease 45 kN
 - E. Increase 45 kN
 - F. Decrease 30 kN

Determine the numerical value (magnitude only) of the change in shear for section $12 < x < 24$ m as well as the behavior (shape) of the shear for the same region.

7. (3 pt) Change = _____ kN (magnitude only)
- ✓ A. 36
 - B. 29
 - C. 61
 - D. 24
 - E. 54
 - F. 30

8. (3 pt) Behavior: 

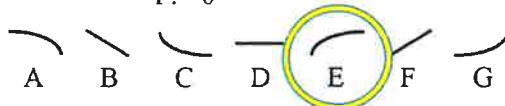
9. (3 pt) What happens to the shear diagram at $x = 9$ m?
- ✓ A. Decrease 30 kN
 - B. Increase 24 kN
 - C. Decrease 24 kN
 - D. Increase 54 kN
 - E. Decrease 54 kN
 - F. Decrease 36 kN



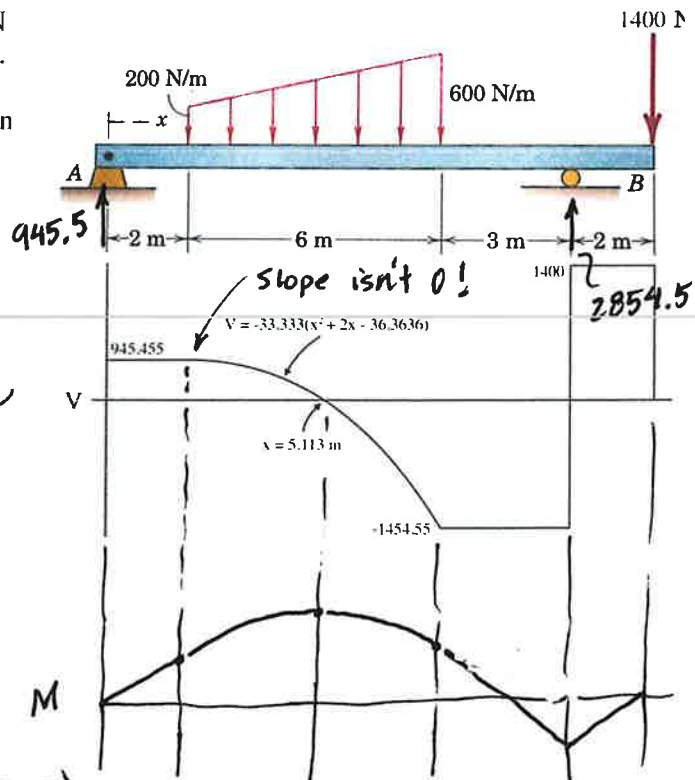
Provide information for constructing the **moment diagram**. The x origin is on the left and the support reactions are $A_y = 945.455$ N acting up, $B_y = 2854.55$ N acting up. A is a pin and B is a roller.

Determine the numerical value (magnitude only) of the change in moment for section $2 < x < 5.113$ m as well as the behavior (shape) for the same region.

10. (3 pt) Change = _____ N-m (magnitude only)
- ✓ A. 1638.9
 - B. 1890.9
 - C. 945.45
 - D. 1966.2
 - E. 4363.6
 - F. 0

11. (3 pt) Behavior: 

12. (3 pt) What happens to the moment diagram at $x = 11$ m?
- ✓ A. No change (slope change!)
 - B. Decrease 200 N-m
 - C. Increase 600 N-m
 - D. Decrease 400 N-m
 - E. Increase 400 N-m
 - F. Decrease 2400 N-m



#10 integrate V on TI-89 !!

$$\int (-33.33 \cdot (x^2 + 2x - 36.364), x, 2, 5.113)$$

$$= \Delta M = 1638.5 \text{ N}\cdot\text{m}$$

Cannot use the parabola formula:

$$A = \frac{2}{3} ab$$

13. (4 pt) Determine the order of the equation for the internal bending moment for the region $8 < x < 11$ m on the beam shown.

- ✓ A. first order, linear
- B. constant
- C. second order, parabolic
- D. zero
- E. third order, cubic

Section 3 - Friction

14. Determine the minimum force P to initiate motion of the uniform 300 N rod AB up the wall. The contact surface at A is smooth, whereas the coefficient of static friction between the rod and the floor at B is $\mu_s = 0.3$

- A. 90 N
- B. 190 N
- ✓ C. 290 N
- D. 300 N
- E. 390 N

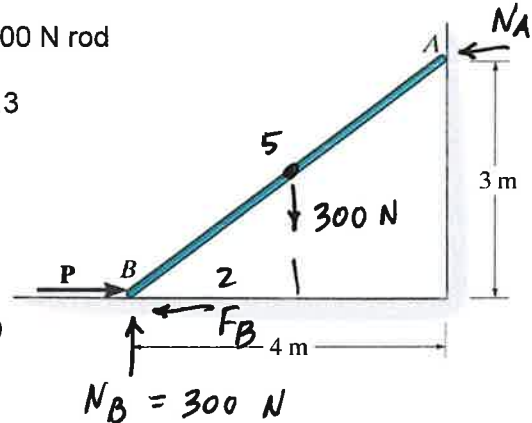
By inspection: $N_B = 300$ N

Then, $F_B = .3(300) = 90$ N

$$+\circlearrowleft \sum M_B = 0; \quad 300(2) - N_A(3) = 0$$

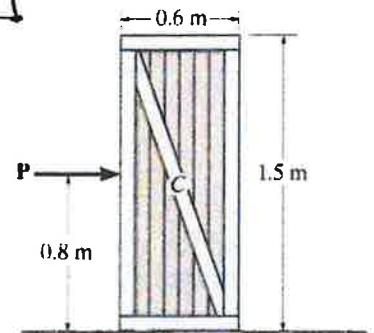
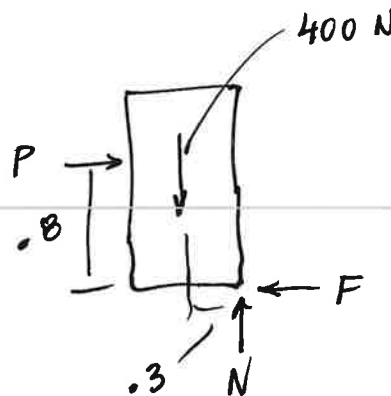
$$N_A = \frac{600}{3} = 200$$

Finally, $\rightarrow \quad P - 90 - 200 = 0; \quad \boxed{P = 290 \text{ N}}$



15. Determine the minimum horizontal force P required to cause the crate to move. The crate weighs 400 N and the coefficient of static friction between the crate and the plane is $\mu_s = 0.4$.

- A. 100 N
- B. 120 N
- C. 130 N
- ✓ D. 150 N
- E. 180 N



Slip: $F = .4(400) = \boxed{160 \text{ N} = P_{\text{slip}}}$

Tip: $+\circlearrowleft \sum M_N = 0; \quad P(.8) - 400(.3) = 0;$

$$\boxed{P = 150 \text{ N}} \sim \text{Will tip first,}$$

16. The coefficient of friction is $\mu = 0.15$ between the fixed posts at B and C and the cord. What is the mass m of D required so that motion of D impends downward?

- A. 60 kg
- B. 70 kg
- C. 80 kg ✓
- D. 90 kg
- E. 100 kg

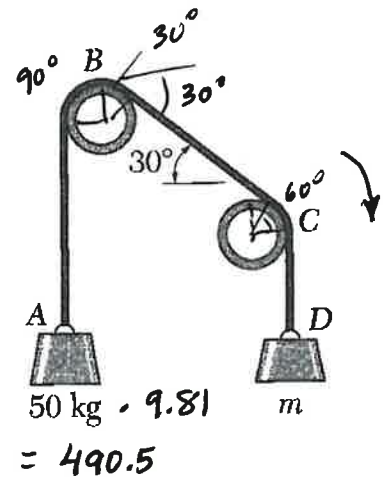
$$\theta = 90 + 30 + 60$$

$$= 180 \cdot \frac{\pi}{180} = \pi \text{ rad}$$

$$T_2 = T_1 e^{\mu \beta}$$

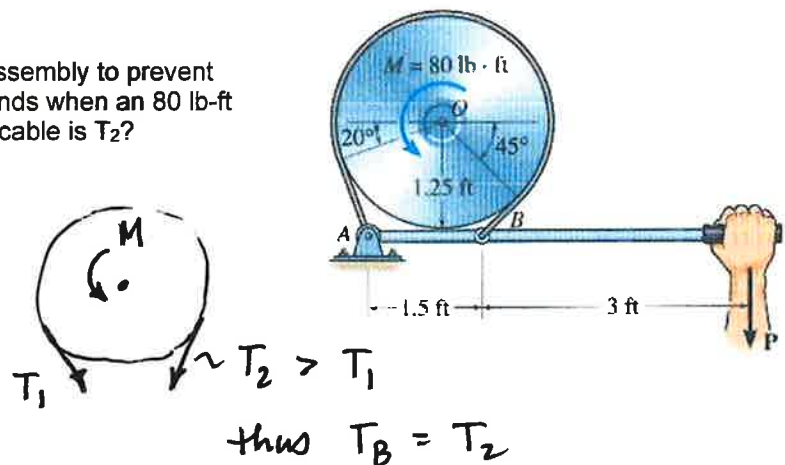
$$= (490.5) e^{(.15)(\pi)}$$

$$= \frac{785.8 \text{ N}}{9.81} = 80.0 \text{ kg}$$



17. Force P is applied to the handle and belt assembly to prevent motion of the drum. If motion of the drum impends when an 80 lb-ft torque is applied to the drum as shown, which cable is T_2 ?

- A. Cable A
- B. Cable B ✓



18. To test if a wedge is self-locking, one does the following:

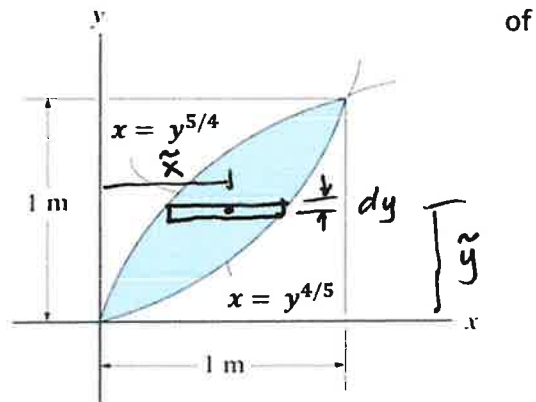
- A. Apply force P pushing on the wedge with $P = 0$.
- ✓ B. Apply force P pulling on the wedge with $P = 0$.
- C. Apply force P on the wedge with $P = 0$ for the raising case.
- D. Apply force P pushing on the wedge with $\mu = 0$.
- E. Apply force P pulling on the wedge with $\mu = 0.5$.

Section 4 – Centroids

19. (5 pts) When using a **horizontal** differential element area to solve for the location of $\bar{x}$ for the shaded area, what expression would be correct for dA ?

- A. $(y^{4/5} - y^{5/4}) dy$ ✓
- B. $(y^{4/5} + y^{5/4}) dy$
- C. $0.5(y^{5/4} - y^{4/5}) dy$
- D. $(y^{5/4} - y^{4/5}) dy$
- E. $0.5(y^{4/5} + y^{5/4}) dy$

$$dA = (y^{4/5} - y^{5/4}) dy$$



20. (5 pts) A ring is generated by rotating the quarter-circular area about the x axis. Determine its volume if the radius $a = 2$ cm.

- A. 135.2 cm^3 ✓
- B. 125.7 cm^3
- C. 153.2 cm^3
- D. 140.1 cm^3
- E. 177.6 cm^3

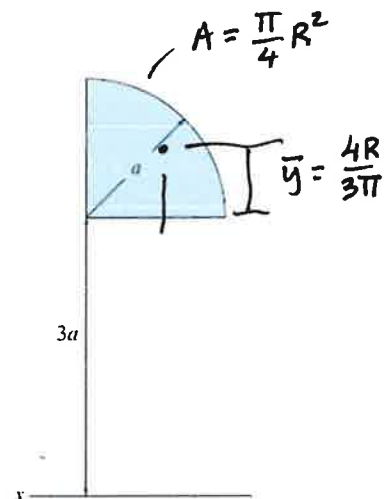
$$V = \sum \theta \bar{r} A$$

$$\theta = 2\pi? \quad 360^\circ?$$

$$V = (2\pi) \left(3(2) + \frac{4(2)}{3\pi} \right) \left(\frac{\pi}{4} (2)^2 \right)$$

$$= 2\pi \left[6 + \frac{8}{3\pi} \right] (\pi)$$

$$V = 135.2 \text{ cm}^3$$



21. (5 pts) A ring is generated by rotating the quarter-circular area about x axis. Determine its surface area if the radius $a = 2$ cm.

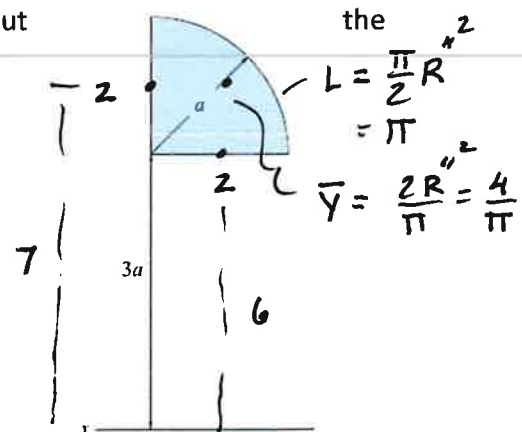
- A. 319.5 cm^2
- ✓ B. 309.6 cm^2 306.9
- C. 297.5 cm^2
- D. 389.7 cm^2
- E. 200.1 cm^2

$$A = \sum \theta \bar{r} L$$

$$= 2\pi \left[6(2) + 7(2) + \left(6 + \frac{4}{\pi} \right) (\pi) \right]$$

$$A = 2\pi [12 + 14 + 7.273\pi]$$

$$A = 306.9 \text{ cm}^2$$



Assume $\theta = 2\pi$

22. (5 pt) Determine the location of $\bar{y}$ for the shaded area.

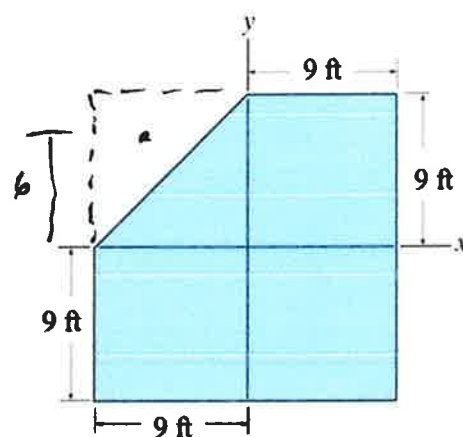
- A. -0.86 ft ✓
- B. -0.57 ft
- C. -0.75 ft
- D. -0.91 ft
- E. -0.68 ft

$$\bar{y} = \frac{\sum \tilde{y} A_i}{\sum A_i}$$

$$\sum A_i = 18(18) - \frac{1}{2}(9)(9) = 283.5$$

$$\bar{y} = \frac{0(18)(18) - 6[40.5]}{283.5}$$

$$\boxed{\bar{y} = -0.857 \text{ ft}}$$



23. (5 pt) Determine the location of $\bar{z}$ for the composite rod if $h = 6$ ft.

- A. 0.97 ft ✓
- B. 0.86 ft
- C. 0.91 ft
- D. 0.78 ft
- E. 0.62 ft

$$\bar{z} = \frac{\sum z_i L_i}{\sum L_i}$$

$$= \frac{3(6) + 0(4\pi)}{6 + 4\pi}$$

$$\boxed{\bar{z} = 0.969 \text{ ft}}$$

